The Effects of Family Planning Program on Gender Equality in Human Capital

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Abstract

From the 1960s through the 1980s, the South Korean government carried out an intense family planning program which played a substantial role in decreasing the fertility rate. In this paper, we estimate the gender-specific effect of the family planning program on educational attainment and occupational outcomes using a difference-in-differences method. The data used are the Population and Housing Census, and records of the achievements of the family planning program. Our findings indicate that an increase in supplies of oral pills reduces the gender gap in educational attainment and possessing a professional career. A key mechanism is that the decreased number of children within a household may alter the investment in each child's education, which affects their decision of having a professional career in the long run. We then discuss potential channels through which the family planning program led to the increased investment in female education.

Keywords: Family Planning, Education, Gender Gap

*Alphabetical order

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1 Introduction

In the mid-twentieth century, global efforts to reduce fertility rate began to rise driven by the fear of post-war population explosion. Many developing countries had a population growth rate as high as 3 percent a year which was two- or threefold the highest rates ever experienced by European countries. International organizations and foundations were established in order to advocate family planning programs, and some East Asian countries and India launched a national family planning program. These efforts turned out to be successful, lowering the world's fertility rate from 5 in 1960 to 2.5 in 2010.[1]

The effectiveness of family planning programs on reducing fertility rate has been supported by studies that investigate cases of various countries. De Silva and Tenreyro (2017) has shown a negative relationship between the intensity of family planning and the change in Total Fertility Rate (TFR) across developing countries. Kearney and Levine (2009) examines the family planning program in the United States and finds that greater eligibility for family planning services reduced overall births to non-teens by about two percent and to teens by over four percent. Bailey (2012) also provides robust evidence that federal family planning programs reduced the US fertility rate between 1964 and 1973. There also have been studies that examine the effects of One-Child-Policy on fertility rate in China (Zhang, 2017; Ahn, 1994; McElroy and Yang, 2000).

South Korea, along with other East Asian countries, was one of the earliest and most successful adopters of family planning programs. In 1960, the TFR in South Korea was 6.0 and it continued to stay at a high level for several years. One reason was that the majority of people were engaged in agriculture at the time and family members constituted a significant part of the labor force. In addition, high infant mortality induced parents to bear extra children to reach the number of children they desire. Although the infant mortality rate decreased as public health improved, it took some time until people actually perceived the fact and changed their child bearing behaviors. Parents kept having extra kids, and the annual population growth rate from 1955 to 1960 was 2.9%,[2] which was much higher than normally expected.

The government realized that population control was inevitable to achieve economic growth, and thus, implemented a family planning program in 1962. Education on contraceptive methods and surgeries were provided and contraceptive tools were distributed to each public health center by the government. As a result, the fertility rate as well as the number of children within a household fell rapidly. According to the quantity-quality trade

[1] World Bank data

[2] KIHASA (2016) reviews the population status before 1960.

off theory, this might have led to more investment per child and a corresponding increase in the child's educational attainment.

We expect the educational attainment for girls to have improved more than it had for boys after the family planning program was implemented. Since son preference was prevalent, it was usually the boys who had the lion's share of the parents' investment. Thus, the marginal benefit of investment in children's education would have been larger for girls. Moreover, introduction of contraceptive tools should have been more effective for girls as it can delay marriage and childbearing which affects their education and labor participation. In fact, the educational attainment has increased sharply in South Korea since the 1960s. This growth was more dramatic for girls than boys, which resulted in a decreased gender gap in education. The ratio of females who attained higher education is 80.5% for cohorts born in 1964 and 98.1% for cohorts born in 1978, while that of male increased from 89.7% to 97.3%. Female and male junior college graduation ratio are 7.6% and 9.7% respectively for the 1964 cohort. Female junior college graduation ratio grew more rapidly and reached 23.9% for the 1978 cohort and exceeded that of male which is 19.5%. Also, the share of female four-year college graduates doubled during 1964-1978, increasing from 17.4% to 34.8%. This narrowed the gender gap by 18.4%p to 10.1%p.

Previous studies support the positive relationship between family planning and investment in female human capital. Huang et al. (2016) examines the effect of the One-Child-Policy(OCP) in China on senior high school completion using subgroup analysis by gender. The fine rate for excess children was used to measure the intensity of the OCP, and their findings indicate that the OCP significantly increased the education level of girls whereas it did not for boys. In line with this work, using a DID approach, Geng (2020) proves that China's OCP had a positive impact on educational attainment of girls in the post-OCP period and that the impact for boys was modest. Miller (2010) further extends the positive impact of family planning on women as to include women's socio-economic status and independence. Goldin and Katz (2002) carefully analyzes the effect of access to birth control pills on marriage decision and how it affects career for women.

In this paper, we try to identify the causal link between the family planning program and gender-specific educational and job outcomes using a difference-in-differences method. In order to use a DID method, we construct a district-level panel data containing information on several educational attainment level - higher education, junior college graduate and four-year college graduate - and the ratio of individuals having a professional job by gender. We also use government supplies of contraceptive tools between 1964 and 1978 to measure the intensity of the family planning program. While previous studies on family planning programs use indirect measures of family planning programs, we use the quantity of oral

pills distributed in each public health center by the government to measure the exposure to the family planning program.

We find strong evidence that the family planning program led to positive educational outcomes, especially for the girls. Our estimates show that on average, an increase of 1%p in the exposure of females to the family planning program measured by oral pills increases the share of more than high school graduate for female by by 7.1%p, graduating junior college by 7.6% and graduating four-year college by 6.3%. In the long run, the ratio of females having a professional job by 13.8%p. Our findings on the occupational outcomes imply that the experience of family planning program has a long-lasting effect for females.

A subgroup analysis on districts without major cities shows that our findings are robust with respect to the change of administrative borders of districts throughout 1964-1978. We also examine the effects of the family planning program at different ages and find that the policy was most effective at the birthyear of children. In addition, we use condoms and I.U.D.s(Intra-uterine Devices) as alternative measures of the intensity of the family planning program for robustness check. Our results suggest that oral pills were most effective among various contraceptive tools and this might be due to the convenience of use for women. We then discuss potential mechanisms such as family size channel, household income channel and weakened son preference channel that can help develop a deeper understanding of these results.

By these exercises, we aim to provide empirical evidence to Becker and Lewis (1973) by measuring the child's quality with educational attainment and draw implications for the developing countries where family planning is viewed as one of the most cost-effective ways to improve human resource and promote gender equality. We also show that in the long run, higher educational attainment increases the quality of female labor supply measured by the ratio of females having a professional job.

The remainder of this paper is organized as follows. Section 2 briefly introduces the background of the family planning program in South Korea during 1962-1980. In section 3, we discuss the data and the identification strategy used to measure the effect of the family planning program on educational attainment. Section 4 presents results and section 5 discusses potential mechanisms that can explain the the results. Section 6 concludes the paper.

2 The Family Planning Program in South Korea

The family planning program in South Korea was implemented in 1962 when the government recognized that reducing fertility rate was inevitable to accomplish economic growth. At

the time, Korea was experiencing an unprecedented population growth as the temporary suspension of the Korean War brought back many refugees to their homeland and made people feel safe. The fertility rate was also at a high level. This not only meant a general lack of resources per person, but also reflected a shortage in educational facilities for children as well. For instance, elementary schools in 1960 contained 80 to 100 children per class, and schools had to operate on morning and afternoon shifts (Yang, 1979). Although there were efforts to promote family planning prior to 1962, few people were aware of these issues and the results were far from meaningful.

A shift began in 1961 as the Planned Parenthood Federation of Korea(PPFK) was established to facilitate more comprehensive and organized approach to family planning. Although the PPFK had to suspend its activities due to the Military Revolution, it succeeded in persuading the new government to adopt a policy that supported family planning. The new government, established as a result of the Military Revolution in 1961, sought to stabilize the country by boosting economic growth and felt the need to restrain population growth that was offsetting the gains made from economic improvements. In 1962, the government officially pronounced that a family planning program was to be included in the First Five-Years' Economic Development Plan(1962-1966) and lifted the ban on production and importation of contraceptives at once. Since then, the family planning program had remained as a crucial part of the government policy throughout the next three Economic Development Plans.

Although each of the Five-Years' Economic Development Plans had some differences such as the target group and specific strategies of family planning program, the key elements of the program remained consistent: providing people with public supported contraceptive methods and educating people about the necessity of family planning. To familiarize people with the concept of family planning, the government first trained and assigned family planning workers in every district so that by 1964, every district had at least two family planning workers. Their job was to motivate couples to practice family planning and issue coupons that enabled them to get free supplies or surgeries at nearby hospitals or clinics, going door to door.

The government aimed to reduce the population growth rate to 25, 20, 15, and 13 per thousand by the end of each Five-Years' Development Planning period(1966, 1971, 1976, and 1981). In order to achieve such a goal, targets for family planning practice rates were set and overall national targets were divided into annual targets by specific control methods which were followed by further detailed allocation to local government units. These methods, which included I.U.D. surgery, sterilization and distribution of condoms at the beginning, were expanded to include oral pills from 1968. The goals and targets have been readjusted from time to time to reflect demographic reality, development of contraceptive technology,

and budget appropriation (Yang, 1979). Aside from these direct methods of family planning, the government made efforts to fight son preference entrenched in the society as well.

These trends in family planning are summarized well in the slogans promoted by the government. In the late 1960s, the popular slogan was "3·3·35", which urged people to have only 3 children with at least 3 years between births, and stop having kids after 35. Then the slogan changed to "Daughter or son, just have two kids and raise them well" in the 1970s and finally, "Even two kids are more than enough" in the 1980s when the TFR hit the population replacement rate 2.1. Nevertheless, the government's will to push for the family planning remained strong and as a result Korea's fertility rate turned to 1.7 in 1995, which was way below the global average, 3.04. It was in 1996 that the government finally shifted its focus from restraining population growth to improving health of the nation. The TFR of South Korea lowered from 6.0 in 1960 to 2.7 in 1978 and this has become one of the most successful examples in the history of family planning around the world. It is commonly believed that this success played a significant role not only in economic growth but in raising investment and attainment in education as well.

3 Data and Empirical Strategy

The district-level panel data used in this study are drawn from three reports published by the Korean Institute for Family Planning and the Population Census 2010 and 2015 2% sample. The cohort born in 1978, the latest cohort born during our sample period, would be age 32 in 2010 so for most of them, education should have been completed by then. Also, data on labor market status of all our sample cohorts would also be available. The Population Census 2005 2% sample was excluded because it does not contain information on the birth region in district-level. Moreover, the latest cohort would be 27 years old in 2005, which is a period for men to search for a job after college since military service is mandatory in South Korea. Thus, the working population of this cohort in 2005 could cause a selection problem.

3.1 Main Variables

The reports that contain the number of oral pills and condoms distributed and the number of I.U.D. and vasectomy surgeries performed in each public health center are *Achievement of the National Family Planning Program by Health Center – based on C.Y.P.* (1975), *Statistics on Population and Family Planning in Korea* (1978) and *1978 Family Planning Program Evaluation* (1979). Since there was only one public health center per each district, all the health centers and the districts can be matched one to one. The amount of supplies for each

contraceptive in 1964-1969, 1970-1977 and 1978 are from each of these reports respectively.

Our sample is restricted to the cohorts born during 1964-1978. Although the family planning program continued until the late 1980s, government supply of contraceptive tools began in 1964 and data are available only from 1964 to 1978. We argue that this does not lead to selection because the proportion of fertile women who accepted contraceptive tools at the early stage is large enough. Also, by the mid-1970s, the influence of family planning program in reducing TFR would not have been significant compared to what it had been in the 1960s.

We also assume that the family planning program would not have had an immediate impact on the decision of childbearing. Rather, it is more reasonable to assume that the number of births in a specific year is affected by the experience of the family planning program in the previous years. To construct an exposure measure of the family planning program for a given year and district, we use the sum of the amount of the contraceptives distributed in the given year and the amount distributed in the past 4 years to represent a 5-year cumulative effect of family planning. This is then divided by the total sum of the number of females aged 15-49 during the same period in the same district for normalization.

Therefore, our measure of how exposed fertile women were to the family planning program is defined as

$$FP\ 5s_j^b = \frac{FP_j^{[b-4,b]}}{P_j^{[b-4,b]}} \tag{1}$$

where $FP_j^{[b-4,b]}$ is the cumulative amount of the contraceptives distributed from year $b-4$ to b in district j. $P_j^{[b-4,b]}$ is the cumulative number of females aged 15-49 from year $b-4$ to b in district j. This can be interpreted as an index representing the degree of exposure to the family planning program for fertile women.

Figure 1 presents the exposure rate of family planning program measured with data available on four distinct contraceptives from 1964 to 1978. Oral pills were introduced in 1968 and the intensity of the family planning program reached its peak in 1975. Although data on various contraceptives are available, we choose the oral pills as our main explanatory variable because oral pills were distributed in the largest quantity. We will discuss results using condom and I.U.D. as alternative measures of the intensity of the family planning program in section 4.3.3. We exclude vasectomy since it was a new technology back then and therefore not commonly used. Indeed, the number of vasectomy shows little variation across districts. The mean of $FP5s$ measured by vasectomy is only 0.0035, which means that for a given five-year period, only 0.35% were experienced the operation of vasectomy. Thus,

$Pil\ 5s$, $Con\ 5s$ and $IUD\ 5s$ each denotes the exposure index of fertile women calculated by using oral pills, condoms and I.U.D.s.

*** Figure 1 here ***

3.2 Contemporary Outcome and Control Variables

We investigate the human capital level of our sample in terms of educational and occupational outcomes. Both of the contemporary outcome variables are drawn from the Population Census 2010 and 2015 2% sample by birth district, province in the census year, sex and birth year. We choose the ratio of entering higher education, junior college graduate ratio and four-year college graduate ratio for males and females as the main outcome variables. Attending middle school is compulsory by law in South Korea. Thus, attending high school means being more educated than what is required. As seen in Figure 2, girls' education level increased faster than that of the boys'.

*** Figure 2 here ***

In order to identify the long-term effect of the family planning program, we use the ratio of individuals with a professional job (hereafter, professional job ratio) for males and females. Job outcomes can be a measure to explore the return to human capital investment in the long run. We define professional jobs as jobs that require human capital investment and high educational attainment. Among the ten Korean Standard Classification of Occupations, professional jobs include managers and professionals.[3] The ratio is enumerated from the sample of labor force participants. Table 1 shows that there was a dramatic increase in the ratio of females participating in the labor market with a professional career.

*** Table 1 here ***

As seen in Figure 2, education level differ across districts. Thus, we include a set of control variables that may affect the educational attainment for the regression on the educational outcomes. We use sex ratio below age 2 as a proxy of the son preference. If parents have strong son preference, investment in their children's education might differ by the child's gender. The logarithm of the population size is included to control for the economic development and urbanization of a district. Literature on human capital have shown

[3]The rest include clerical, service and sales workers, skilled agricultural, forestry and fishery workers, craft and related trades workers, plant and machine operators and assemblers, elementary occupations and armed forces occupations.

strong influence of parents' educational level on children's educational attainment. The educational level of parent cohorts in a district is measured with a dummy variable indicating whether males and females aged 20-55 have graduated from primary school.[4] We measure this respectively for males and females because characteristics of mothers and fathers may have different impacts on the performances of the child (Hwang et al., 2019). These variables are derived from the Population Census 1966, 1970, 1975 and 1980 2% sample.

We also control birth year trend by including birth year, the quadratic and the third polynomial of it. We further include two variables to control for the education supply. The number of high schools per a hundred high school students and the number of teachers per student in high schools indicate the quantity and quality of education supply respectively. These education supply variables are expected to control the effects of various policy changes regarding education. The data are from the *Statistical Yearbooks* of each district published in 1960, 1965, 1970, 1975 and 1980. The years in between are interpolated by the adjacent years. Summary statistics of the control variables are given in Table 2.

*** Table 2 here ***

Similar control variables are used for the analyses with the occupational outcomes as a dependent variable, but these include age and the quadratic of age instead of birth year. The four-year college completion ratio is added since we assume that higher educational attainment should lead to a higher chance of having a professional job, which can provide evidence for the long term effect of the family planning program. Control variables of the educational supply are excluded when the professional job ratio is the outcome variable.

3.3 Difference-in-Differences Method

We estimate the following equation.

$$Y_{p,j}^{t,b,g} = \beta_1 FP\ 5s_j^b \times Female_{p,j}^{t,b} + \beta_2 FP\ 5s_j^b + \beta_3 Female_{p,j}^{t,b} + X_j^b + \delta_j + \delta_p + \delta_b + \delta_t + \epsilon_{p,j}^{t,b} \tag{2}$$

where $Y_{s,j}^{t,b,g}$ is the share of each level of educational attainment or the professional job ratio of the male or female cohort born in year b in district j. p denotes the province of residence in census year t and g denotes the gender of the cohort. $FP\ 5s_j^b$ is the exposure index of the family planning program for the cohort born in year b in district j. We use this as a

[4]For the parent generation in the period 1964-1978, compulsory education in South Korea encouraged students to attend primary school.

proxy of the intensity of the family planning program. Note that the data is in district level, so the dependent variable consists of ratios of contemporary educational and occupational levels for boys and girls separately. $Female_{s,j}^{t,b}$ is a dummy variable that equals 1 if $Y_{s,j}^{t,b}$ is a contemporary outcome variable for females. X_j^b indicates the set of control variables, and δ_j, δ_s, δ_b, δ_t each denotes the fixed effect for district of birth, province of residence in the census year, birth year and census year.

The coefficient of the interaction term, β_1, measures the additional impact of the family planning program for girls relative to that for boys. We expect β_1 to be positive, which implies that the family planning program had a stronger impact of increasing the educational attainment on girls than it did on boys. The data includes birth cohorts born during 1964 to 1978 and covers 168 birth districts and 8 current provinces. In order to account for the potential correlation of observations within the same cluster, we report standard errors clustered at district and birth year levels.

4 Results

4.1 Main results

Table 3 - Table 5 present the regression results from estimating equation (2) using oral pills to measure the intensity of the family planning program. Column (1) presents the result of the regression with contemporary province, census year, birth district, birth year and the interaction of birth year and census year fixed effects. Column (2) includes covariates, and column (3) is the result of the identical regression in column (2), but with the restricted sample excluding those with missing values for supplies of education. Column (4) includes the additional control variables of educational supplies and column (5) shows the result with all control variables including birth year trend. The negative coefficients of $Female$ suggest that females had lower educational attainment. However, it is clearly shown that oral pills induced a greater improvement in all three levels of educational attainment for girls than boys, which is reflected in the positive and statistically significant coefficients of $Pil5s \times Female$. This is consistent with our expectation and general findings in population education literature.

*** Table 3 here ***

*** Table 4 here ***

*** Table 5 here ***

The estimated coefficient of the interaction term between $Pil\ 5s$ and $Female$ in Table 3 is 0.071, which means that a unit increase in the exposure index of oral pills induces an additional increase of 7.1%p in high school completion ratio for girls when compared to the increase in high school completion ratio for boys. Similarly, a unit increase in the exposure index of oral pills increased junior college and four-year college completion ratio for girls by 7.6%p and 6.3%p respectively (see col. 5 of Table 4 and 5).

As in other countries, junior colleges in South Korea focus on practical education and vocational training and a major reason for entering a junior college is to get a job. Therefore, it can be inferred from the fast growth in the number of female junior college graduates that more girls expected to participate in the labor force and the introduction of oral pills must have played a role. Graduating a four-year college not only implies a higher probability of getting a job, but also getting a higher quality job. In this respect, it can be concluded that the positive effect of oral pills on four-year college completion ratio for girls reduced the gender gap in human capital.

4.2 Long-term Effect: Occupational Outcomes

The regression results with the ratio of professional job holders among those in labor market participation as the outcome variable are provided in Table 6. The coefficient of the interaction term is 13.9%p (col. 5). Although not presented in this paper, the coefficient of the interaction term is 2.3%p when using the professional job ratio obtained from the total population sample by sex, and the results are robust.

These findings indicate that the increase in the share of females with a professional career can be explained by the family planning program to some extent. This is important in the sense that the family planning program has a long lasting effect on human capital and labor market outcomes. The results suggest that the family planning program not only changed the investment of the parents but also the career decision of the children in the long run.

*** Table 6 here ***

4.3 Robustness Checks

4.3.1 Subgroup Analysis without Major Cities

We conduct a subgroup analysis without major cities because of two reasons. First, the administrative borders of the capital city, Seoul, and metropolitan cities[5] have changed in a great degree. Most of the metropolitan cities were originally a part of a province,

[5]Metropolitan cities include Busan, Incheon, Daegu, Gwangju, Daejeon and Ulsan.

but as these cities grew bigger, they absorbed nearby regions and became a distinct city. When constructing the district level panel data, we incorporated the districts that have gone through an administrative change into the largest administrative district that includes them.[6] Since districts in the major cities have gone through more administrative changes than the others, it might not be proper to compare these cities with other districts.

Second, major cities may be a special case since they are urban areas and went through a dramatic development in economic and cultural aspects including the population growth. The coefficients are presented in Table 7 and this shows that our findings are robust. The estimated coefficients on the interaction term between $Pil\ 5s$ and $Female$ are 0.097, 0.191 and 0.103 for high school, junior college and four-year college graduate ratio respectively.

*** Table 7 here ***

4.3.2 Effects at different ages

The results above show the effect of the family planning program at the year of birth for 1964-1978 cohorts. The effects of the family planning program could have been heterogeneous depending on age at which the children were exposed to the family planning. For example, the effects of the introduction of oral pills could be different for children who were just born and for those who entered elementary school in that year. In order to examine this, we conduct a parallel analysis, this time regressing the same outcome variables on the explanatory variables with a lag of six or twelve years. This is due to the fact that children enter elementary school at age six and middle school at age twelve in South Korea.

Columns (1)-(3) in Table 8 present the effect of the the family planning program at age six. Since the index of the exposure to the family planning program is available for 1964 to 1978, the birth cohorts included in the sample for these three regressions are from 1958 to 1972. Similarly, columns (4)-(6) show the effect of the family planning program when the 1952-1966 birth cohorts were at age of twelve. It is hard to compare the coefficients of interest with those in Table 3-5 for the regressions have been done with a different set of cohorts. Nonetheless, given the mean value of the dependent variables for distinct samples, the coefficients of the interaction term suggest that the effect of the family planning program was most effective at the birth year of the children.

*** Table 8 here ***

[6]For example, Daejeon used to be a district belonging to Chungcheongnam-do, but it became a metropolitan city with the integration with Daeduk-gun, which was also a district in Chuncheongnam-do. We consider Daejeon and Daeduk-gu as Daejeon-si in Chungcheongnam-do.

4.3.3 Alternative measures

We perform similar analyses with the data on condoms and I.U.D. surgeries per each public health center and report the results in Table 9 and 10. Our data allows us to compare the effects of various contraceptive tools, which is a distinctive feature that have not yet been presented in previous literature. The coefficients of interest for condoms and I.U.D.s are positive and significant when the outcome variable is the higher education ratio. However, for junior college and four-year college completion, the coefficients are not significant.

Interestingly, our results suggest that oral pills are most effective among the contraceptives introduced during the family planning program. This could be due to the fact that pills are relatively easy to use and can be controlled by females. They also incur less pain and less health-related risks (Birdsall and Chester, 1987). The negative coefficient of the *Female* term indicates that females have lower education level than males in general.

*** Table 9 here ***

*** Table 10 here ***

5 Discussions

Section 4 reveals that the family planning program have improved the educational attainment for girls while the boys have not been affected as much. However, it remains a black box how or why the decreased sibling size translates into differences in the improvement of educational attainment between girls and boys. In this section, we explore potential mechanisms that have been suggested throughout various literature and provide evidence that can validate or invalidate the channels. We discuss family size channel, household income channel and weakened son preference channel in particular, and suggest the first two mechanisms as the most plausible channels. They have been presented by Bailey (2013), which focuses on the educational outcome of children in general, not on the gender gap. Therefore, we provide additional explanation on how these channels can be used to account for the decrease in the gender gap.

5.1 Family Size Channel

The family planning program had the effect of reducing the number of children within a household. Fewer children in a household means an increase in the availability of parental time and material resources per child, increasing the educational attainment of children in

general. However, as seen in the high school completion ratio of boys and girls at the beginning of the program, boys were getting sufficient investment for completing high school compared to girls. In addition, college admission required significantly more resource and investment while the chance of entering college was low. These altogether made marginal returns on education of girls higher relative to marginal returns on education of boys. Therefore, parents would have allocated extra resources induced by the family planning more towards the education of girls. This argument is based on the assumption that parents gain utility from the educational outcome of their children, viewing educational spending by parents as a form of investment good. This can then be understood as risk diversification by parents as suggested by Geng (2020).

Geng (2020) also presents a mechanism for a case where educational spending can be viewed as a consumption good. Even if parents gain utility from spending on their children's education regardless of the outcome, a smaller number of children means a lower cost of raising children and this still generates an income effect. For households where daughters were underinvested while sons were not due to a lack of resources, parents would have increased the investment in daughters when the availability of the resources increased.

5.2 Household Income Channel

Aside from generating an income effect through the reduction of the number of children, the family planning program can actually increase the income level of households. The family planning program provided couples with cheaper and more reliable contraceptive methods. This reduced the immediate and expected costs of delaying childbearing, freeing up more resources to be invested in the parents' own human capital. Delaying parenthood would have allowed parents to obtain more education, work experience and job training, which is likely to increase their lifetime earnings. This is supported by various literature that have been covered in the introduction of this paper such as Goldin and Katz (2002) and Miller (2010). They show that earlier access to the pill increased women's investment in their careers and their socio-economic status including wages. Subsequently, this increase in household income would have prompted more investment in girls for similar reasons as the family size channel.

5.3 Weakened Son Preference

As mentioned in section 2, weakening son preference was another target of the family planning program. If the family planning program not only reduced the TFR but also weakened son preference, investment decisions of the parents in children's education would have become less gender-specific. In other words, the educational attainment of girls would have

increased due to the change in their parents' perceptions of gender norms. However, a survey taken in 1977 from Kim (1991) shows that the percentage of parents who wished their daughters to graduate from college was 34% whereas 54% wanted their sons to be a college graduate. This indicates that parents had different expectations depending on the child's gender.

If this was the main channel, educational attainment of females in districts with high son preference would have improved more. To test this hypothesis, we conduct a parallel regression with subgroups by districts with high son preference and districts with moderate son preference. In order to divide districts into these subgroups, it is essential to note the following two characteristics of son preference in South Korea.

First, son preference could only be revealed after the 1980s. Son preference has long been a convention in South Korea due to the influence of Confucianism, but it could not be reflected in the sex ratio at birth due to the lack of technology - it was impossible to know the sex of a child in advance to delivery. However, after the introduction of ultrasound technology which brought about sex-selective abortion, sex ratio at birth soared during the 1980s. In 1990, the sex ratio at birth reached its peak with 117 boys being born per 100 girls.

Second, although sex ratio at birth seems to vary across regions, rankings change little over time. Hong et al. (2018) documents how the strength of son preference may differ by regions. During the *Joseon* dynasty, the concentration of *Seowon*, which is a private Confucian academy, may have played a roll in forming intense son preference in regions with Seowons. Regions that recorded strong son preference are *Daegu, Gyeongbuk, Ulsan, Gyeongnam* and *Busan* and the ranking does not seem to have changed much.

Using these facts, we use a triple-differences model to further test the hypothesis that the boost in female education level can be attributed to weakened son preference. We assume that the ranking of districts by sex ratio at birth remains stable over time and divide the districts into two groups - districts with abnormally high sex ratio at birth and districts with natural sex ratio at birth. In general, natural sex ratio at birth is considered to be between 104 and 107. Using the Population Census 1990 and 1995 2% sample, when son preference could be revealed by sex ratio at birth, we calculate the the sex ratio of children below age 2 and classify a district as having strong son preference if the sex ratio at birth is higher than 107. We define the variable $High\ SR$ to be 1 for the districts with abnormally high sex ratio at birth.

The results in Table 11 show that girls in districts with high level of son preference did not experience a noticeable improvement in educational attainment compared to the girls in the other group. The coefficients are not significant after including control variables. Although

this analysis cannot entirely deny the impact of lessened son preference, we conclude that weakening son preference was not a main channel through which the family planning closed the gender gap in high school completion rates.

*** Table 11 here ***

6 Conclusion

This paper shows that the intensity of the family planning program, proxied by the supply of contraceptives per fertile women for five years, had a positive effect on enhancing various levels of educational attainment for girls. We first begin by noticing that South Korea's rapid decrease in education gap coincided with the period of national family planning. South Korea implemented intense family planning policies as a way of accelerating economic growth, and has seen a dramatic decline in TFR since then. The quantity-quality trade off theory suggests that this may have led to more investment per child, particularly for girls since they were being significantly under-invested by their parents.

Previous studies have shown that China's OCP led to a decrease in the gender gap in education. Exploiting the district-by-year variation in the government supply of contraceptive tools, we also identify the effect of family planning on gender gap in education. Through difference-in-differences method, we find that a unit increase in the exposure to oral pills led to increase in more than high school completion ratio of females relative to males by 0.071, junior college completion ratio by 0.076 and four-year college completion ratio by 0.063. These effects were stronger when children were at the year of birth, which indicates that the effect of family planning program is more effective at the early stage of children.

The most plausible explanation behind these results is that the family planning program generated an income effect through the reduction of family size and increase in parents' income, which allowed extra resources to be invested in female education. Delaying marriage or childbearing leads to increased household income through parents' investment in their own human capital, while reduced number of children lowers the total cost of raising the children. These extra resources were then directed towards girls as they had higher marginal returns to education compared to the boys.

Our work not only adds to the evidence of the quantity-quality trade off, but also presents insight for developing countries that are pursuing family planning as a way of promoting gender equality. In further research, it would be interesting to explore whether the family planning program affected outcomes of the girls other than educational attainment and labor market decisions.

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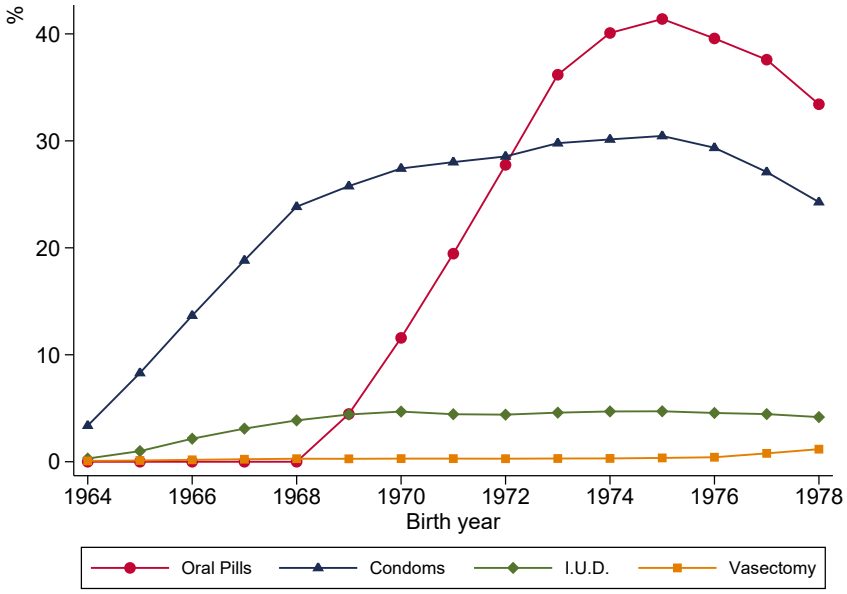

Figure 1: Exposure Rate to the Family Planning Program

Notes. Authors' calculations from records of the achievement of the family planning program. The exposure rate is defined as in equation 1 with 168 districts.

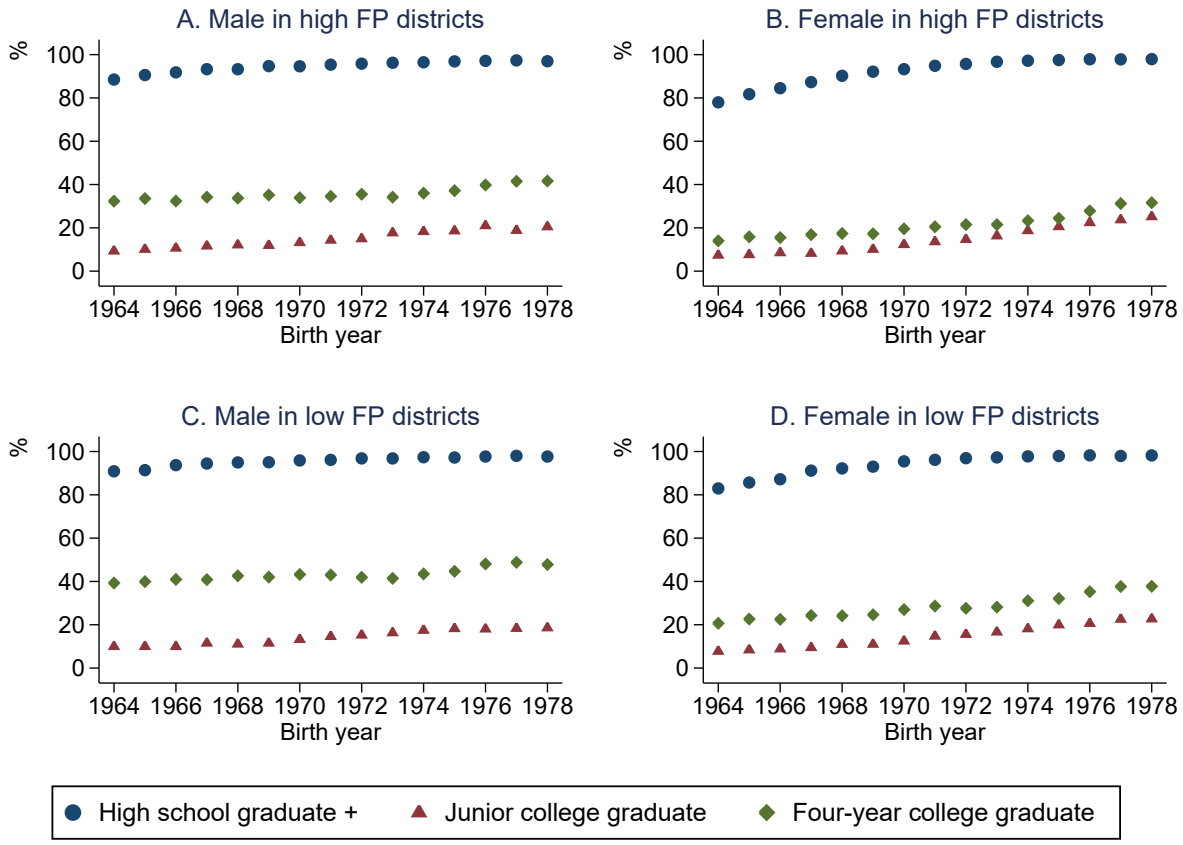

Figure 2: Education level by gender and exposure to the family planning program

Notes. Authors' calculations from 2010 and 2015 2% sample. Districts with high FP are districts that are over the median of *Pil* 5*s* and districts with low FP are districts that have *Pil* 5*s* that are under the median of it.

Table 1: Professional job ratio (%) by gender and exposure to the family planning program

Birth year	Total			High FP districts			Low FP districts		
	Male	Female	Diff.	Male	Female	Diff.	Male	Female	Diff.
1964	22.6	16.4	6.3	20.6	14.0	6.5	24.7	18.7	6.0
1965	22.2	17.3	4.9	20.0	15.5	4.5	24.3	19.1	5.2
1966	22.3	18.1	4.2	19.3	16.0	3.3	25.2	20.2	5.0
1967	22.5	19.8	2.6	19.3	17.8	1.5	25.5	21.9	3.7
1968	23.4	20.7	2.7	20.4	18.5	1.9	26.2	22.8	3.4
1969	24.4	20.8	3.6	22.1	18.8	3.3	26.7	22.7	4.0
1970	23.8	23.2	0.7	20.9	21.0	-0.1	26.6	25.3	1.3
1971	25.0	25.8	-0.7	22.0	23.8	-1.7	27.8	27.7	0.2
1972	24.7	26.3	-1.6	23.0	23.7	-0.7	26.2	28.7	-2.5
1973	25.0	27.3	-2.3	22.3	25.2	-2.9	27.5	29.2	-1.7
1974	26.4	29.0	-2.6	24.5	27.2	-2.7	28.1	30.7	-2.6
1975	26.1	31.7	-5.7	23.6	29.2	-5.5	28.3	34.1	-5.8
1976	27.5	33.0	-5.4	26.1	31.1	-5.0	28.8	34.6	-5.8
1977	26.8	34.9	-8.2	24.6	33.7	-9.1	28.8	36.0	-7.2
1978	26.2	35.2	-8.9	24.4	33.3	-8.9	27.9	36.8	-8.9

Notes. Districts with high FP are districts that are over the median of $Pil\ 5s$ and districts with low FP are districts that have $Pil\ 5s$ that are under the median of it. Diff. is the subtraction of Male (%) and Female (%).

Table 2: Summary statistics of control variables by exposure to the family planning program

Covariates	Total		High FP districts		Low FP districts	
	Mean	SD	Mean	SD	Mean	SD
Sex ratio (age <2)	112.49	18.58	113	21.02	111.98	15.75
log(population)	12	0.6	11.69	0.4	12.3	0.6
Father: primary graduate	0.82	0.11	0.78	0.09	0.85	0.11
Mother: primary graduate	0.66	0.16	0.61	0.13	0.7	0.17
No. of high schools per 100	0.29	0.32	0.37	0.35	0.23	0.28
Teachers per student	0.04	0.01	0.04	0.02	0.04	0.01

Notes. Sex ratio (age <2), log(population), Father: primary graduate and Mother: primary graduate are drawn from Census 2% sample of 1966, 1970, 1975 and 1980. No. of high shools per 100 and Teacher per student are from the Statistical Yearbooks.

Table 3: Effect of oral pills on more than high school completion ratio

Dependent variable:	**More than high school completion ratio**				
	(1)	(2)	(3)	(4)	(5)
Pil 5s x Female	0.098***	0.099***	0.071***	0.071***	0.071***
	(0.023)	(0.023)	(0.018)	(0.018)	(0.006)
Female	-0.028***	-0.028***	-0.017**	-0.017**	-0.017***
	(0.008)	(0.008)	(0.006)	(0.006)	(0.003)
Pil 5s	0.052***	-0.006	0.007	0.006	-0.028***
	(0.013)	(0.013)	(0.011)	(0.011)	(0.007)
Sex raio (age<2)		-0.000**	-0.000*	-0.000*	-0.000**
		(0.000)	(0.000)	(0.000)	(0.000)
log(population)		-0.022***	-0.010**	-0.012***	-0.017***
		(0.005)	(0.003)	(0.003)	(0.003)
Father: primary graduate		0.005	0.038	0.034	0.064
		(0.051)	(0.051)	(0.050)	(0.052)
Mother: primary graduate		0.093**	0.105**	0.106***	0.107***
		(0.039)	(0.036)	(0.035)	(0.038)
No. of high schools per 100				0.000	0.000
				(0.000)	(0.000)
Teachers per student				-0.123**	-0.109**
				(0.057)	(0.051)
FE	Y	Y	Y	Y	Y
Birth Year Trend	N	N	N	N	Y
Dependent Variable Mean	0.95	0.95	0.96	0.96	0.96
Observations	136,776	133,765	91,339	91,339	91,339

Notes. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Birth Year Trend means controlling birth year, birth year ^{2} and birth year ^{3}. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Effect of oral pills on junior college completion ratio

Dependent variable:	**Junior college completion ratio**				
	(1)	(2)	(3)	(4)	(5)
Pil 5s x Female	0.084***	0.084***	0.076***	0.076***	0.076***
	(0.019)	(0.019)	(0.020)	(0.020)	(0.006)
Female	-0.008*	-0.008*	-0.004	-0.004	-0.004
	(0.004)	(0.004)	(0.005)	(0.005)	(0.002)
Pil 5s	0.044***	-0.026*	-0.036**	-0.035**	-0.013*
	(0.012)	(0.014)	(0.015)	(0.014)	(0.007)
Sex raio (age<2)		-0.000	-0.000	-0.000	-0.000
		(0.000)	(0.000)	(0.000)	(0.000)
log(population)		-0.019**	-0.020***	-0.018***	-0.016***
		(0.007)	(0.005)	(0.006)	(0.004)
Father: primary graduate		0.109**	0.041	0.040	0.017
		(0.038)	(0.045)	(0.045)	(0.042)
Mother: primary graduate		0.045	0.118***	0.112***	0.117***
		(0.027)	(0.037)	(0.037)	(0.031)
No. of high schools per 100				-0.000*	-0.000
				(0.000)	(0.000)
Teachers per student				-0.108	-0.117*
				(0.074)	(0.065)
FE	Y	Y	Y	Y	Y
Birth Year Trend	N	N	N	N	Y
Dependent Variable Mean	0.15	0.15	0.15	0.15	0.15
Observations	136,776	133,765	91,339	91,339	91,339

Notes. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Birth Year Trend means controlling birth year, birth year ^{2} and birth year ^{3}. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Effect of oral pills on four-year college completion ratio

Dependent variable:	**Four-year college completion ratio**				
	(1)	(2)	(3)	(4)	(5)
Pil 5s x Female	0.084***	0.084***	0.063**	0.063**	0.063***
	(0.025)	(0.025)	(0.026)	(0.026)	(0.011)
Female	-0.138***	-0.138***	-0.127***	-0.127***	-0.127***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.005)
Pil 5s	-0.014	-0.042**	-0.015	-0.019	-0.041***
	(0.020)	(0.016)	(0.017)	(0.019)	(0.012)
Sex raio (age<2)		0.000	0.000	0.000	0.000
		(0.000)	(0.000)	(0.000)	(0.000)
log(population)		0.019**	0.023***	0.014**	0.010*
		(0.006)	(0.008)	(0.005)	(0.006)
Father: primary graduate		0.080	0.130**	0.124*	0.140**
		(0.049)	(0.059)	(0.059)	(0.059)
Mother: primary graduate		0.133***	0.077	0.096*	0.104**
		(0.042)	(0.055)	(0.052)	(0.049)
No. of high schools per 100				0.001	0.001*
				(0.001)	(0.001)
Teachers per student				0.095	0.115
				(0.095)	(0.089)
FE	Y	Y	Y	Y	Y
Birth Year Trend	N	N	N	N	Y
Dependent Variable Mean	0.35	0.35	0.36	0.36	0.36
Observations	136,776	133,765	91,339	91,339	91,339

Notes. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Birth Year Trend means controlling birth year, birth year ^{2} and birth year ^{3}. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Effect of oral pills on professional job ratio

Dependent variable:	**Professional job ratio**				
	(1)	(2)	(3)	(4)	(5)
Pil 5s x Female	0.175***	0.139***	0.132***	0.138***	0.139***
	(0.006)	(0.005)	(0.007)	(0.005)	(0.005)
Female	-0.016***	0.043***	0.044***	0.043***	0.043***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Pil 5s	0.120***	-0.025***	-0.026***	-0.015**	-0.021***
	(0.006)	(0.006)	(0.008)	(0.006)	(0.006)
Age		-0.002	0.001	-0.002	-0.002
		(0.002)	(0.002)	(0.002)	(0.002)
Age2		-0.000	-0.000***	-0.000	-0.000
		(0.000)	(0.000)	(0.000)	(0.000)
Four-year college graduate		0.427***	0.422***	0.424***	0.424***
		(0.004)	(0.005)	(0.004)	(0.004)
Sex raio (age<2)				0.000	0.000
				(0.000)	(0.000)
log(population)				0.011***	0.015***
				(0.003)	(0.004)
Father: primary graduate					0.051**
					(0.021)
Mother: primary graduate					-0.002
					(0.019)
FE	Y	Y	Y	Y	Y
Dependent Variable Mean	0.27	0.27	0.28	0.27	0.27
Observations	127,674	127,674	85,002	124,872	124,872

Notes. Fixed effects include census year, province in the census year and birth district. Sampling weights are applied. Standard errors are clustered at birth district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Effect of oral pills on education, without major cities

Dependent variable:	**High School**		**Junior college**		**Four-year college**	
	(1)	(2)	(3)	(4)	(5)	(6)
Pil 5s x Female	0.098***	0.097***	0.088***	0.088***	0.103***	0.103***
	(0.018)	(0.004)	(0.018)	(0.006)	(0.018)	(0.008)
Female	-0.030***	-0.030***	-0.013***	-0.013***	-0.150***	-0.150***
	(0.007)	(0.002)	(0.003)	(0.002)	(0.005)	(0.003)
Pil 5s	-0.013	-0.048***	-0.039**	-0.024***	-0.078***	-0.071***
	(0.015)	(0.008)	(0.015)	(0.008)	(0.016)	(0.010)
Sex raio (age<2)	-0.000	-0.000*	-0.000	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
log(population)	-0.009	-0.015**	-0.027***	-0.025***	0.023***	0.022***
	(0.006)	(0.006)	(0.008)	(0.006)	(0.007)	(0.006)
Father: primary graduate	0.016	0.022	0.045	0.039	0.083	0.070
	(0.057)	(0.061)	(0.048)	(0.046)	(0.050)	(0.054)
Mother: primary graduate	0.061	0.047	0.079*	0.090**	-0.002	0.008
	(0.042)	(0.043)	(0.040)	(0.036)	(0.050)	(0.050)
No. of high schools per 100	-0.000	-0.001	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Teachers per student	-0.120**	-0.098**	-0.114	-0.115	0.145*	0.132*
	(0.050)	(0.046)	(0.080)	(0.071)	(0.081)	(0.078)
FE	Y	Y	Y	Y	Y	Y
Birth Year Trend	N	Y	N	Y	N	Y
Dependent Variable Mean	0.95	0.95	0.14	0.14	0.32	0.32
Observations	75,744	75,744	75,744	75,744	75,744	75,744

Notes. Major cities include the capital city and the metropolitan cities - Seoul, Busan, Daejeon, Ulsan, Incheon, Gwangju and Deagu. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Effect of oral pills on education and occupation outcomes at age 6 or 12

Dependent variable:	**At age 6**			**At age 12**		
	High school (1)	Junior college (2)	Four-year college (3)	High school (4)	Junior college (5)	Four-year college (6)
Pil 5s x Female	0.192***	0.019***	0.051***	0.133***	-0.012***	-0.052***
	(0.017)	(0.005)	(0.009)	(0.024)	(0.004)	(0.007)
Female	-0.089***	-0.018***	-0.156***	-0.156***	-0.019***	-0.146***
	(0.008)	(0.002)	(0.004)	(0.011)	(0.001)	(0.003)
Pil 5s	-0.052***	-0.025***	-0.077***	0.002	0.014***	0.067***
	(0.016)	(0.006)	(0.010)	(0.017)	(0.004)	(0.008)
Sex raio (age<2)	-0.000	-0.000	0.000*	-0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
log(population)	-0.054***	-0.009***	-0.001	-0.060***	0.002	0.018**
	(0.008)	(0.003)	(0.006)	(0.010)	(0.003)	(0.008)
Father: primary graduate	0.135	0.040	-0.082*	0.363***	0.024	-0.034
	(0.107)	(0.026)	(0.048)	(0.088)	(0.018)	(0.059)
Mother: primary graduate	0.335***	0.013	0.244***	0.292***	-0.025*	-0.028
	(0.083)	(0.021)	(0.048)	(0.076)	(0.015)	(0.046)
No. of high schools per 100	0.001**	0.000***	0.000	0.001**	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Teachers per student	-0.238*	-0.052*	0.185*	-0.300***	0.015	0.154*
	(0.136)	(0.030)	(0.094)	(0.097)	(0.032)	(0.080)
FE	Y	Y	Y	Y	Y	Y
Birth Year Trend	Y	Y	Y	Y	Y	Y
Dependent Variable Mean	0.88	0.10	0.29	0.76	0.07	0.23
Observations	93,937	93,937	93,937	92,313	92,313	92,313

Notes. Fixed effects include census year, province in the census year and birth district. Sampling weights are applied. Standard errors are clustered at birth district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Effect of condoms on education

Dependent variable:	**High School**		**Junior college**		**Four-year college**	
	(1)	(2)	(3)	(4)	(5)	(6)
Con 5s x Female	0.069**	0.069***	0.019	0.019*	0.020	0.019
	(0.029)	(0.015)	(0.026)	(0.011)	(0.043)	(0.025)
Female	-0.021**	-0.021***	0.004	0.004	-0.122***	-0.122***
	(0.009)	(0.005)	(0.008)	(0.003)	(0.014)	(0.008)
Con 5s	0.020	0.021	-0.031	-0.034*	0.014	0.025
	(0.015)	(0.013)	(0.021)	(0.017)	(0.032)	(0.027)
Sex raio (age<2)	-0.000**	-0.000**	-0.000	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
log(population)	-0.009**	-0.010***	-0.022***	-0.023***	0.016**	0.017**
	(0.003)	(0.003)	(0.006)	(0.004)	(0.006)	(0.007)
Father: primary graduate	0.076	0.076	0.039	0.024	0.139**	0.141**
	(0.048)	(0.051)	(0.044)	(0.044)	(0.059)	(0.059)
Mother: primary graduate	0.095**	0.092**	0.121***	0.141***	0.090	0.086*
	(0.036)	(0.038)	(0.036)	(0.033)	(0.052)	(0.050)
No. of high schools per 100	0.000	0.000	-0.000	-0.000	0.001	0.001*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Teachers per student	-0.126**	-0.119**	-0.106	-0.124*	0.095	0.116
	(0.058)	(0.053)	(0.073)	(0.066)	(0.094)	(0.089)
FE	Y	Y	Y	Y	Y	Y
Birth Year Trend	N	Y	N	Y	N	Y
Dependent Variable Mean	0.96	0.96	0.15	0.15	0.36	0.36
Observations	91,339	91,339	91,339	91,339	91,339	91,339

Notes. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Effect of I.U.D.s on education

Dependent variable:	**High School**		**Junior college**		**Four-year college**	
	(1)	(2)	(3)	(4)	(5)	(6)
IUD 5s x Female	0.491**	0.490***	0.125	0.124	-0.058	-0.061
	(0.217)	(0.119)	(0.194)	(0.078)	(0.344)	(0.178)
Female	-0.023*	-0.023***	0.004	0.004	-0.115***	-0.115***
	(0.011)	(0.006)	(0.009)	(0.003)	(0.017)	(0.009)
IUD 5s	0.399**	0.331***	0.037	-0.201*	0.131	0.333**
	(0.147)	(0.096)	(0.141)	(0.115)	(0.243)	(0.158)
Sex raio (age<2)	-0.000**	-0.000**	-0.000	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
log(population)	-0.008**	-0.010***	-0.017***	-0.021***	0.014**	0.016**
	(0.003)	(0.003)	(0.005)	(0.004)	(0.006)	(0.006)
Father: primary graduate	0.047	0.041	0.041	0.034	0.132**	0.123**
	(0.048)	(0.050)	(0.047)	(0.046)	(0.059)	(0.059)
Mother: primary graduate	0.091**	0.100***	0.109***	0.136***	0.095*	0.091*
	(0.034)	(0.036)	(0.035)	(0.033)	(0.052)	(0.050)
No. of high schools per 100	0.000	0.000	-0.000*	-0.000	0.001	0.001*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Teachers per student	-0.125**	-0.132**	-0.107	-0.123*	0.096	0.110
	(0.056)	(0.054)	(0.074)	(0.068)	(0.094)	(0.089)
FE	Y	Y	Y	Y	Y	Y
Birth Year Trend	N	Y	N	Y	N	Y
Dependent Variable Mean	0.96	0.96	0.15	0.15	0.36	0.36
Observations	91,339	91,339	91,339	91,339	91,339	91,339

Notes. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Effect of Oral Pills on Education by Sex Ratio at Birth

Dependent variable:	**High School**		**Junior college**		**Four-year college**	
	(1)	(2)	(3)	(4)	(5)	(6)
Pil 5s x Female x High SR	-0.022*	-0.022*	-0.006	-0.006	-0.001	-0.001
	(0.012)	(0.012)	(0.012)	(0.013)	(0.020)	(0.022)
Pil 5s x Female	0.080***	0.080***	0.080***	0.079***	0.064**	0.064***
	(0.019)	(0.007)	(0.020)	(0.007)	(0.026)	(0.013)
Female x High SR	0.007	0.007	0.007	0.007	0.005	0.005
	(0.005)	(0.005)	(0.004)	(0.004)	(0.009)	(0.010)
Pil 5s x High SR	0.000	0.000	0.002	0.002	-0.026	-0.027
	(0.011)	(0.011)	(0.014)	(0.013)	(0.016)	(0.017)
Female	-0.020***	-0.020***	-0.007	-0.007**	-0.130***	-0.130***
	(0.007)	(0.003)	(0.004)	(0.003)	(0.009)	(0.006)
Pil 5s	0.007	-0.027***	-0.036*	-0.014	-0.010	-0.033**
	(0.012)	(0.008)	(0.017)	(0.009)	(0.018)	(0.013)
log(population)	-0.009**	-0.015***	-0.018***	-0.015***	0.014**	0.010*
	(0.004)	(0.004)	(0.006)	(0.004)	(0.005)	(0.006)
Father: primary graduate	0.043	0.073	0.043	0.019	0.118*	0.135**
	(0.050)	(0.052)	(0.044)	(0.040)	(0.058)	(0.058)
Mother: primary graduate	0.098**	0.099***	0.110***	0.115***	0.103*	0.110**
	(0.034)	(0.038)	(0.036)	(0.031)	(0.051)	(0.049)
No. of high schools per 100	0.000	0.000	-0.000*	-0.000	0.001	0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Teachers per student	-0.130**	-0.117**	-0.110	-0.119*	0.099	0.119
	(0.061)	(0.054)	(0.076)	(0.067)	(0.097)	(0.091)
FE	Y	Y	Y	Y	Y	Y
Birth Year Trend	N	Y	N	Y	N	Y
Dependent Variable Mean	0.96	0.96	0.15	0.15	0.36	0.36
Observations	91,339	91,339	91,339	91,339	91,339	91,339

Notes. High SR stands for High Sex Ratio. Districts with high sex ratio at birth are the districts that had an abnormally high sex ratio at birth (higher than 107) drawn from the Population Census 1990 and 1995 2% sample. Fixed effects include census year, province in the census year, birth district, birth year, birth year × census year. Sampling weights are applied. Standard errors are clustered at birth district and birth year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$